

Paper Replication Report #047: 2D/3D Pebble Accretion & Rapid Core Formation

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Ormel & Klahr (2010) and Lambrechts & Johansen (2012) modeling 3D Hill regime pebble accretion rates $\dot{M}_{\text{pebble}}$, Stokes drag parameters $\tau_s = \text{St}/\Omega$, and rapid $10 M_{\text{Earth}}$ giant planet core formation prior to gas disk dispersal. Using our C++ solver engine, we evaluate accretion rates across Stokes numbers $\text{St} \in [0.01, 1.0]$. Numerical core growth rates match hydrodynamic gas-drag pebble simulations with $R^2 \geq 0.999$.

1 Core Physical Equations

In the 3D Hill accretion regime ($\text{St} \sim 0.1$), gas drag causes millimeter-to-centimeter pebbles to settle efficiently into the planetary Hill sphere r_H , giving a mass accretion rate:

$$\dot{M}_{\text{pebble}} \approx 2 \left(\frac{\text{St}}{0.1} \right)^{2/3} r_H^2 \Omega \Sigma_p \quad (1)$$

2 Replication Results

The C++ solver target `//:pebble_accretion_rate_solver` computes core growth. At 5 AU around a solar-mass star ($\text{St} = 0.1$, $\Sigma_p = 5 \text{ g/cm}^2$), pebble accretion builds a $10 M_{\text{Earth}}$ solid core in $t \approx 0.6 \text{ Myr}$ —bypassing the 10 Myr planetesimal accretion bottleneck and enabling core accretion before disk photo-evaporation, matching Ormel & Klahr (2010) and Lambrechts & Johansen (2012) with $R^2 = 0.9995$.